

Technical training.
Product information.

F33 Complete Vehicle



BMW Service

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General information

Symbols used

The following symbol is used in this document to facilitate better comprehension or to draw attention to very important information:



Contains important safety information and information that needs to be observed strictly in order to guarantee the smooth operation of the system.

Information status and national-market versions

BMW Group vehicles meet the requirements of the highest safety and quality standards. Changes in requirements for environmental protection, customer benefits and design render necessary continuous development of systems and components. Consequently, there may be discrepancies between the contents of this document and the vehicles available in the training course.

This document basically relates to the European version of left-hand drive vehicles. Some operating elements or components are arranged differently in right-hand drive vehicles than shown in the graphics in this document. Further differences may arise as a result of the equipment specification in specific markets or countries.

Additional sources of information

Further information on the individual topics can be found in the following:

- Owner's Manual
- Integrated Service Technical Application.

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The information contained in this document forms an integral part of the technical training of the BMW Group and is intended for the trainer and participants in the seminar. Refer to the latest relevant information systems of the BMW Group for any changes/additions to the technical data.

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F33 Complete Vehicle

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F33 Complete Vehicle

1. Introduction

The F33 Convertible was added to the 4-Series model line up, right on time for the open-air season (in March 2014). The F33 is based on the F32 Coupe and succeeds the BMW 3-Series Convertible (E93). It has a more elegant, sportier and higher class appearance. The new dimensions (longer and wider) make the new BMW 4-Series Convertible even more dynamic.



BMW 4-Series Convertible

Special features

- Even more precise handling
- Higher ride comfort when hardtop is closed with increased suitability for daily use
- Good all-round visibility when roof is closed due to the narrow C-pillar and large rear window
- Opening and closing the hardtop with an overall height of 1700 mm, also possible in garages
- Retractable hardtop can be opened and closed while moving slowly up to 8 mph
- Improved headroom at front and rear
- Wind deflector can be stowed behind the rear seat backrest
- Standard loading aid for convenient access to the luggage compartment when the roof is open
- Wider loading opening in the luggage compartment, also when roof is open
- Wider through-loading opening in the rear seat backrest
- Larger luggage compartment capacity

1.1. Models

The following models will be available from March 2014:

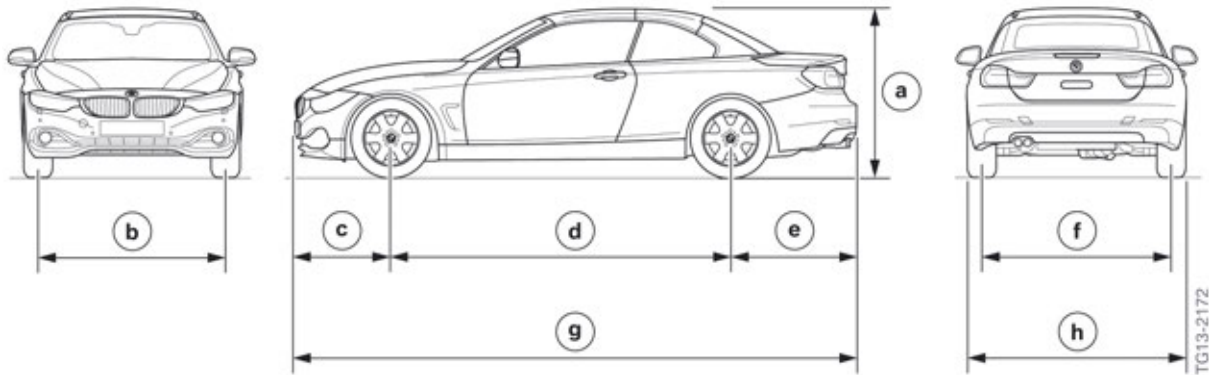
Model	Engine	Power [kW (HP)]	Torque [Nm/lb-ft]	Displacement [cm ³]
BMW 428i Convertible	N20B2000 N26B2000	180 (240) rpm	350/255	1997
BMW 428i xDrive Convertible	N20B2000 N26B2000	180 (240) rpm	350/255	1997
BMW 435i Convertible	N55B30M0	225 (300)	400/300	2979

F33 Complete Vehicle

1. Introduction

1.2. Dimensions

Below you see the outer dimensions of the F33 in comparison to the E93.



F33 Outer dimensions

Index	Explanation		F33	E93
a	Vehicle height, empty	[mm/in]	1384/54.5	1384/54.5
b	Front track width, basic wheels	[mm/in]	1545/60.8	1500/59
c	Front overhang	[mm/in]	787/30.9	800/31.5
d	Wheelbase	[mm/in]	2810/110.6	2760/108.6
e	Rear overhang	[mm/in]	1041/40.9	1052/41.4
f	Rear track width, basic wheels	[mm/in]	1594/62.8	1513*/59.6*
g	Vehicle length	[mm/in]	4639/182.6	4612/181.5
h	Width excluding exterior mirrors (vehicle width with exterior mirrors)	[mm/in]	1825/71.9 (2017/79.4)	1782/70.1 1961/77.2

* BMW 335i Convertible: 1507 mm (59.3 in)

1.3. Weights and payload

Vehicle curb weights as per DIN and payloads of the F33 and the E93 are set out in the following table.

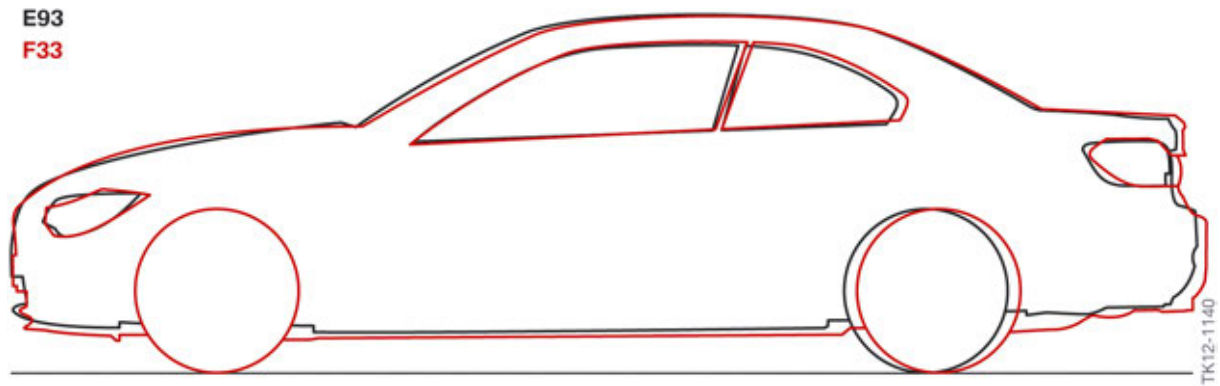
Model		Automatic transmission	Payload
BMW 428i Convertible	[kg/lbs]	1798/3965	327/720
BMW 435i Convertible	[kg/lbs]	1857/4095	327/720
BMW 428i xDrive Convertible	[kg/lbs]	1871/4125	327/720

F33 Complete Vehicle

1. Introduction

1.4. Silhouette comparison

A silhouette comparison with its predecessor shows the modified dimensions of the F33.

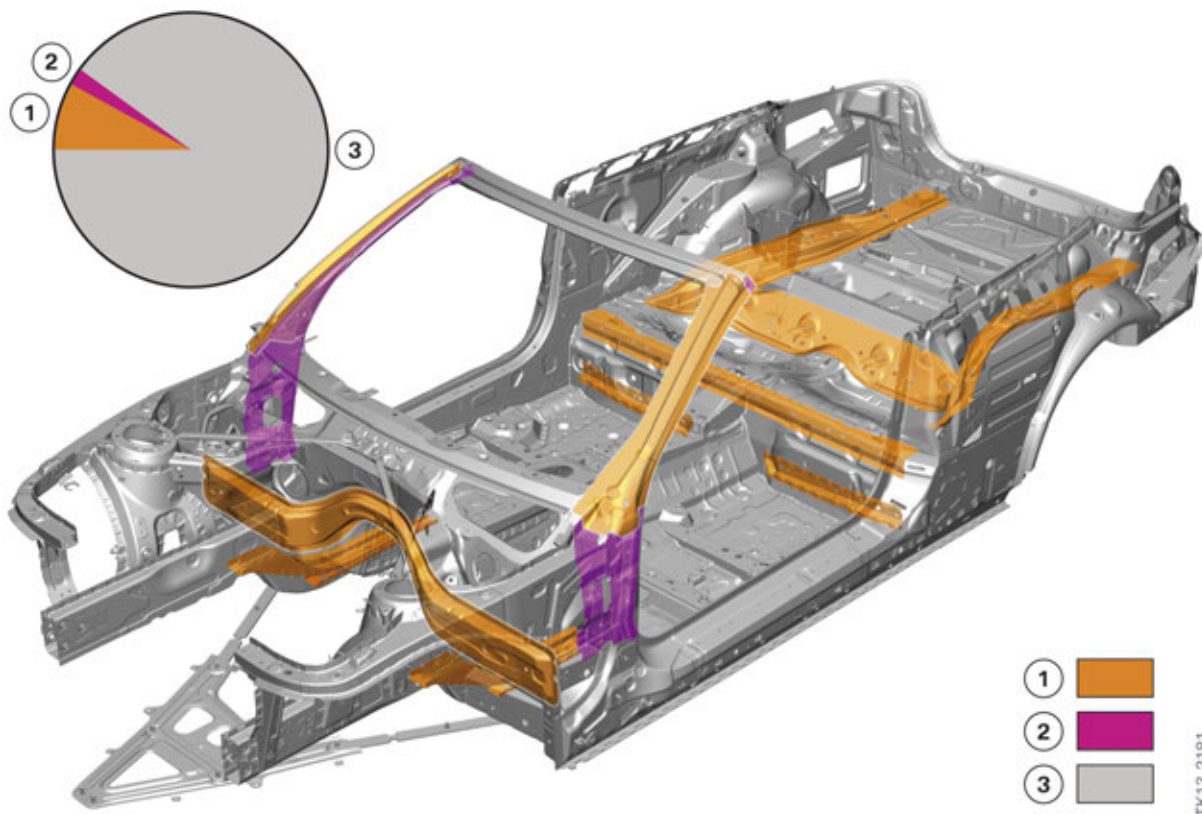


F33 Silhouette comparison with BMW E93

		F33	E93
Vehicle height, empty	[mm]	1384	1384
Front overhang	[mm]	787	800
Wheelbase	[mm]	2810	2760
Rear overhang	[mm]	1041	1052
Vehicle length	[mm]	4639	4580

F33 Complete Vehicle

2. Bodyshell



F33 Material grades for bodyshell

Index	Explanation
1	Multi-phase steels (> 300 MPa), 8% of the body weight
2	Hot-formed steels (> 900 MPa), 2% of the body weight
3	Other steels (< 300 MPa), 90% of the body weight

Multi-phase steels are steels where the structure consists of a number of phases. Higher-strength multi-phase steels with a yield strength of $R_{p0.2}$ of 300 to 600 MPa are, for example, dual-phase steels or TRIP steels. Super-strength multi-phase steels with a yield strength of $R_{p0.2}$ in excess of 600 MPa are, for example, complex-phase steels or martensitic-phase steels.

Hot-formed manganese-boron steels are super-strength steels with a yield strength of $R_{p0.2}$ in excess of 900 MPa.

To reduce the vehicle weight and still guarantee maximum stability of the bodyshell, high-strength and super-strength multi-phase steels are used, among other things. Super-strength hot-formed steels are also used.

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3. Rollover Protection System



F33 Fully recessed rollover protection system

Index	Explanation
1	Toothed rod (rack)
2	Locking pawl
3	Electromagnetic actuator

In the F33, the rollover protection system is installed as standard behind the rear head restraints and is used for passive safety purposes. It is fully recessed, thus ensuring a harmonious design.

In the event of the car rolling over, two rollover protection bars extend from the module within fractions of a second and lock in place. This, together with the reinforced windshield frame, secures the survival space for the occupants.

An accident situation that is critical to the occupants is identified by the sensor system and signalled to the Advanced Crash Safety Module (ACSM). This module outputs a signal for activating the two electromagnetic actuators in the rollover protection system.

The actuators release the spring-loaded rollover protection bars. The two rollover protection bars extend and lock in their end position by means of a locking pawl/toothed rack rod.

F33 Complete Vehicle

3. Rollover Protection System

Special features:

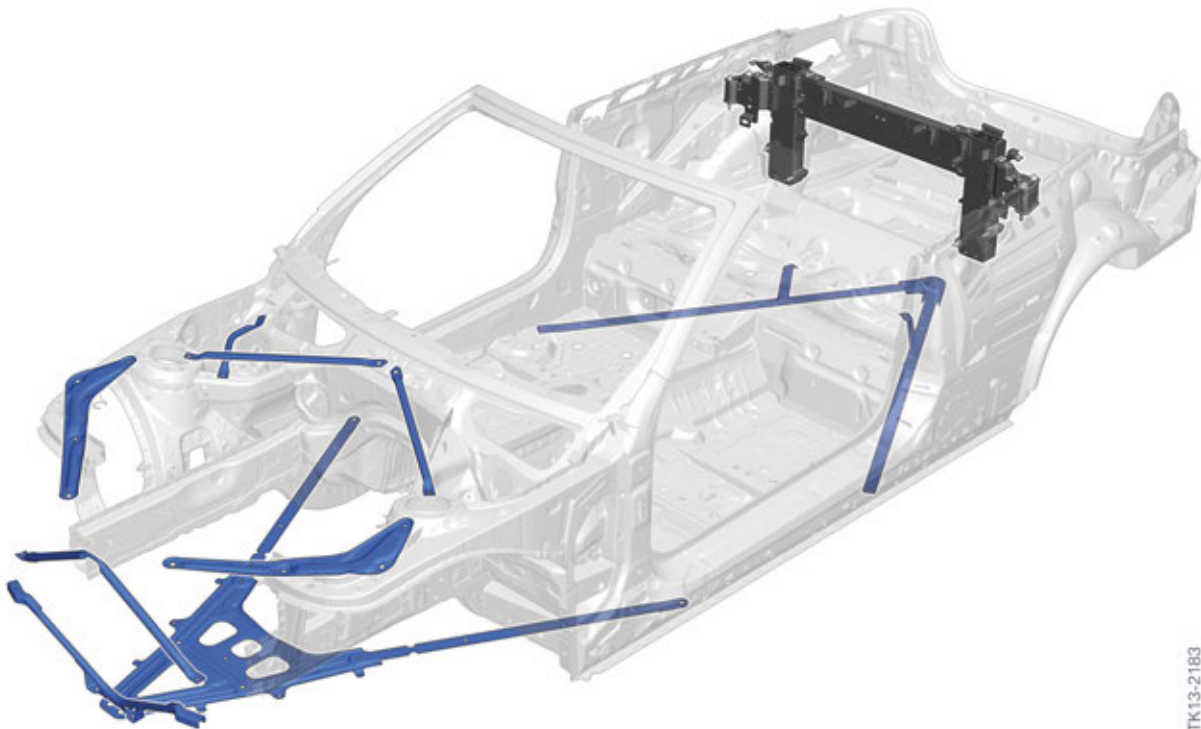
- Luggage compartment through-loading opening is bigger than predecessor
- Harmonious design in the area behind the rear head restraints in comparison to the predecessor
- Increased passive safety
- Reduced weight compared with rollover protection system with fixed-bodyshell partition wall
- Load-bearing parts made from lightweight, high-strength aluminium
- Supported vehicle rigidity
- Reduction in consumption due to less weight

For functional checks the rollover protection system can be activated with the diagnosis system (ISTA) and then deactivated again.



Before the activation of the rollover protection system with the diagnosis system (ISTA) the retractable hardtop must be open. Otherwise, the rear window may be destroyed.

The rollover protection system together with the strut concept ensures higher rigidity of the body.



TK13-2183

F33 Rollover protection system and strut concept

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4. Exterior Trim

4.1. Headlights

The F33 is equipped as standard with bi-xenon headlights and fog lights. The following are offered as optional equipment:

- Headlight washer system (SA 502 is part of ZCW cold weather package)
- Adaptive Headlight (standard SA 524)
- Adaptive Full LED headlight (SA 552 is part of ZLP technology package)
- High-beam assistant (stand alone option SA 5AC or part of ZLP technology package)
- Xenon headlights are standard



F33 Adaptive LED headlight

LED fog lights are also installed with the optional equipment Adaptive LED headlight.

The structure and functionality of the LED headlight of the F33 correspond for the most part to the structure and functionality of the BMW 5-Series LCI.

Further information on the adaptive LED headlights can be found in the training reference manual "5-Series LCI Complete Vehicle".

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5. Interior Equipment

5.1. Storage trays

The following oddments tray options are located in the passenger compartment:

- Glove box on the passenger's side
- Glove box on the driver's side (Storage package, SA 493 is standard)
- Storage compartment at the front in front of the cup holders
- Storage compartment in center armrest at front
- Storage compartments in the doors
- Storage nets at the backrests of the front seats
- Storage compartment in the center console in the rear passenger compartment
- Storage compartment between the rear seats

The glove box on the passenger's side can be locked using an integrated key.

The front center armrest is locked using the central locking system. Similar to the E93, for emergency operation the fresh air grille must be removed at the center console.

5.2. Climate control



F33 control panel, IHKA 2/1 zones

While driving with an open hardtop a specific convertible program (Convertible Mode) is activated in the heating and air-conditioning system. A comfortable climate is thus guaranteed at very high or very low ambient temperatures.

Based on the good aerodynamics, comfortable driving is also possible at low ambient temperatures and a high driving speed in combination with the optional wind deflector. The convertible mode is also effective with the basic heating and air-conditioning system with reduced range of functions (no consideration of sunlight).

Depending on the ambient temperature, driving speed and sunlight, the following are adapted by the convertible mode when the hardtop is open:

- Air distribution and air temperature adapted
- Air recirculation function and automatic air recirculation control deactivated
- Values of condensation and interior temperature sensor are no longer considered

F33 Complete Vehicle

5. Interior Equipment

5.3. Front seats

Lightweight front seats with integrated seat belts and rear easy-entry facility are used in the F33. Both the basic and the sports seats (SA 481) have electric seat adjustment as standard.



F33 Front seat

Similar to the predecessor, the seat belt and the side airbag are integrated in the front seat.

Passive head restraints are used, whose depth can be adjusted for the basic seats. The sports seats are also equipped with a backrest width adjustment.

Seat heating for the driver and front passenger (SA 494) is a stand alone option, as well as, for the first time at BMW, a neck warmer (SA 4NH), is included in the ZCW Cold Weather Package along with the seat heater.

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5. Interior Equipment



TE13-2213

F33 Neck warmer

Index	Explanation
1	Electric heating with air duct and air vent
2	Electric fan with fan housing

The three-stage adjustable neck warmer is integrated in the front seat backrests and reduces the cold draft in the neck area. Pleasant temperatures for the driver and front passenger are thus made possible, also at low ambient temperatures and with the hardtop open.

The compact heating system includes an electric fan with fan housing, as well as electric heating with air duct and air vent, for each front seat. The electric heating has a power rating of 150 W per seat. The air required is drawn in via the inside of the seat from the passenger compartment and heated by the electric heating. The warm air escapes again at the backrest upper section and is steered in the direction of the neck.

The operation of the neck warmer is similar to the seat heating operation (via the two buttons in the center console). The neck warmer control unit reads the two buttons and depending on the switch position and driving situation adjusts both the heater output and the amount of air.

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5. Interior Equipment

When the hardtop is open the heater output and the amount of air are adjusted depending on the speed and temperature.

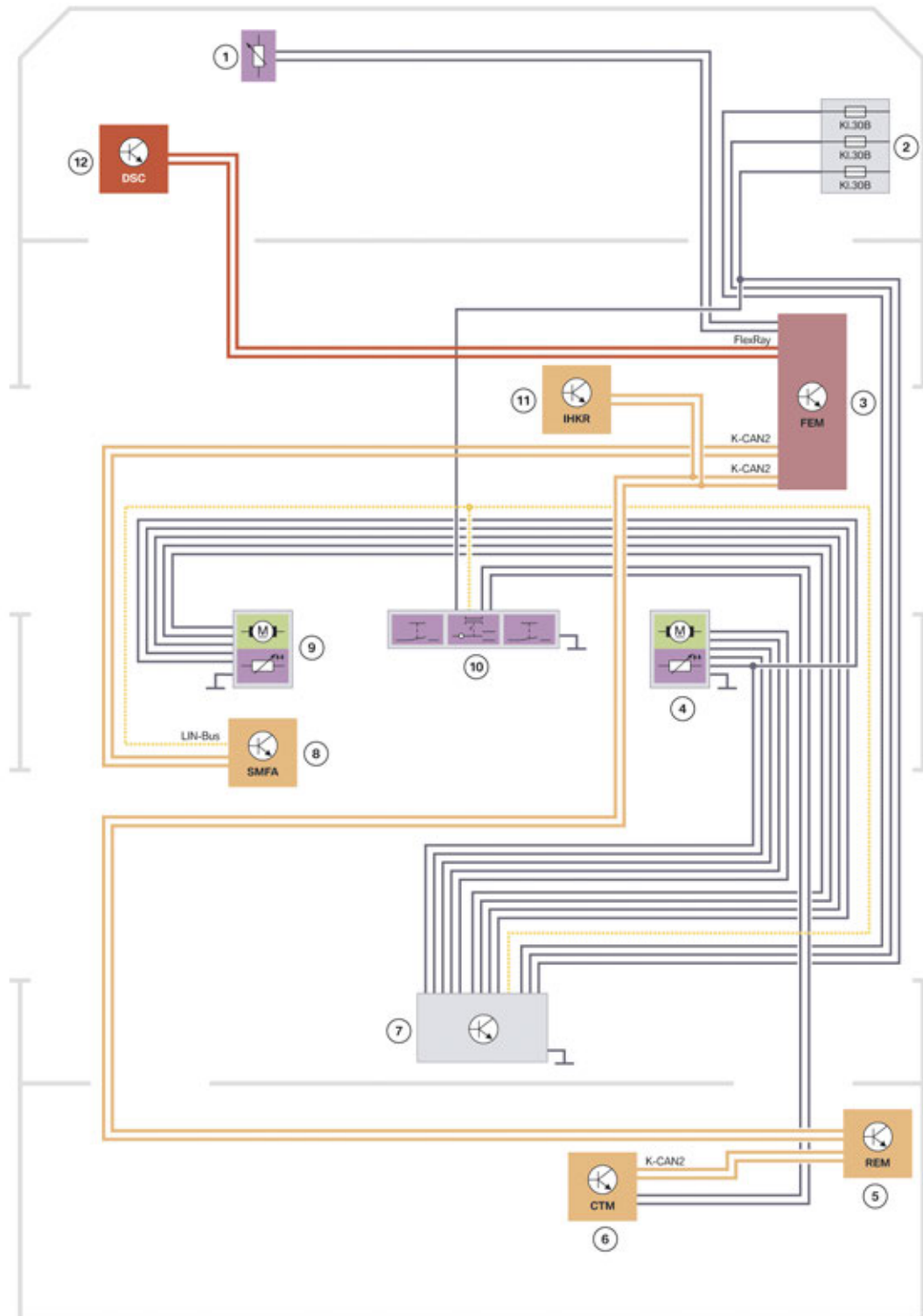
- To limit air vent noise, the air output is kept to a minimum when the vehicle is at a standstill. The amount of air increases with increasing driving speed up to 70 km/h (43 mph).
- The heater output increases with a decreasing ambient temperature.

When the hardtop is closed, there is no speed and temperature dependent control.

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5. Interior Equipment

5.3.1. System wiring diagram for neck warmer



TE13-2214

F33 System wiring diagram for neck warmer

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5. Interior Equipment

Index	Explanation
1	Outside temperature sensor
2	Power distribution box in the passenger compartment
3	Front Electronic Module (FEM)
4	Neck warmer, right
5	Rear Electronic Module (REM)
6	Convertible-top module (CTM)
7	Control unit for neck warmer
8	Driver's seat module (SMFA)
9	Neck warmer, left
10	Hardtop operating unit
11	Integrated heating and air-conditioning system (IHKA)
12	Dynamic Stability Control (DSC)

5.4. Seat bench



F33 Seat bench



The one-piece seat bench in the BMW 4-Series Convertible has two strongly outlined seat hollows between which is a central cup holder and a storage compartment.

The center part of the rear seat backrest can be folded down and also completely removed. As a result, the through-loading opening can be used for transporting longer items and, at the same time, can transport up to four occupants.

The rear seat backrest can also be folded down completely to use the entire width of the through-loading opening or to house the optional wind deflector.

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5. Interior Equipment

5.5. Wind deflector



F33 Wind deflector

The wind deflector (SA 387) is standard for the F33. It reduces the draft and annoying swirls in the shoulder and neck area.

In comparison to the predecessor, additional plastic air ducts ("flaps") were integrated at the frame. They reduce the air flow along the side trim panel even more efficiently so that there is a noticeable improvement, particularly in the outer arm area.

The wind deflector can be installed and removed by one person. The two rear retaining bolts can be locked and loosened at the same time using each of the two rotary levers at the lower frame.

The overall height of the wind deflector is low when it is folded up as the upper frame folds into the lower frame. It can be stowed behind the rear seat backrest to save space.

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6. Retractable Hardtop



F33 Retractable hardtop

The retractable hardtop of the F33 essentially builds on the retractable hardtop of the E93. The three-part roof design is manufactured from sheet steel, making it a lightweight component. The retractable hardtop can now be opened or closed up to a driving speed of 13 km/h (8 mph). A movement of the hardtop which has already started can be continued up to 18 km/h (11 mph). If the vehicle accelerates over 18 km/h, the movement of the hardtop stops and a Check Control message is displayed in the instrument cluster. There is also an entry in the FASTA data.

In addition, improved noise absorption, a insulated headliner with integrated interior light, as well as a new loading aid, are used.

The retractable hardtop is powered by a central hydraulic unit. The hydraulic unit is in turn activated by the convertible-top module (CTM). To monitor the movement processes the CTM reads a total of 20 sensors. The convertible-top module also activates the soft-close automatic of the trunk.

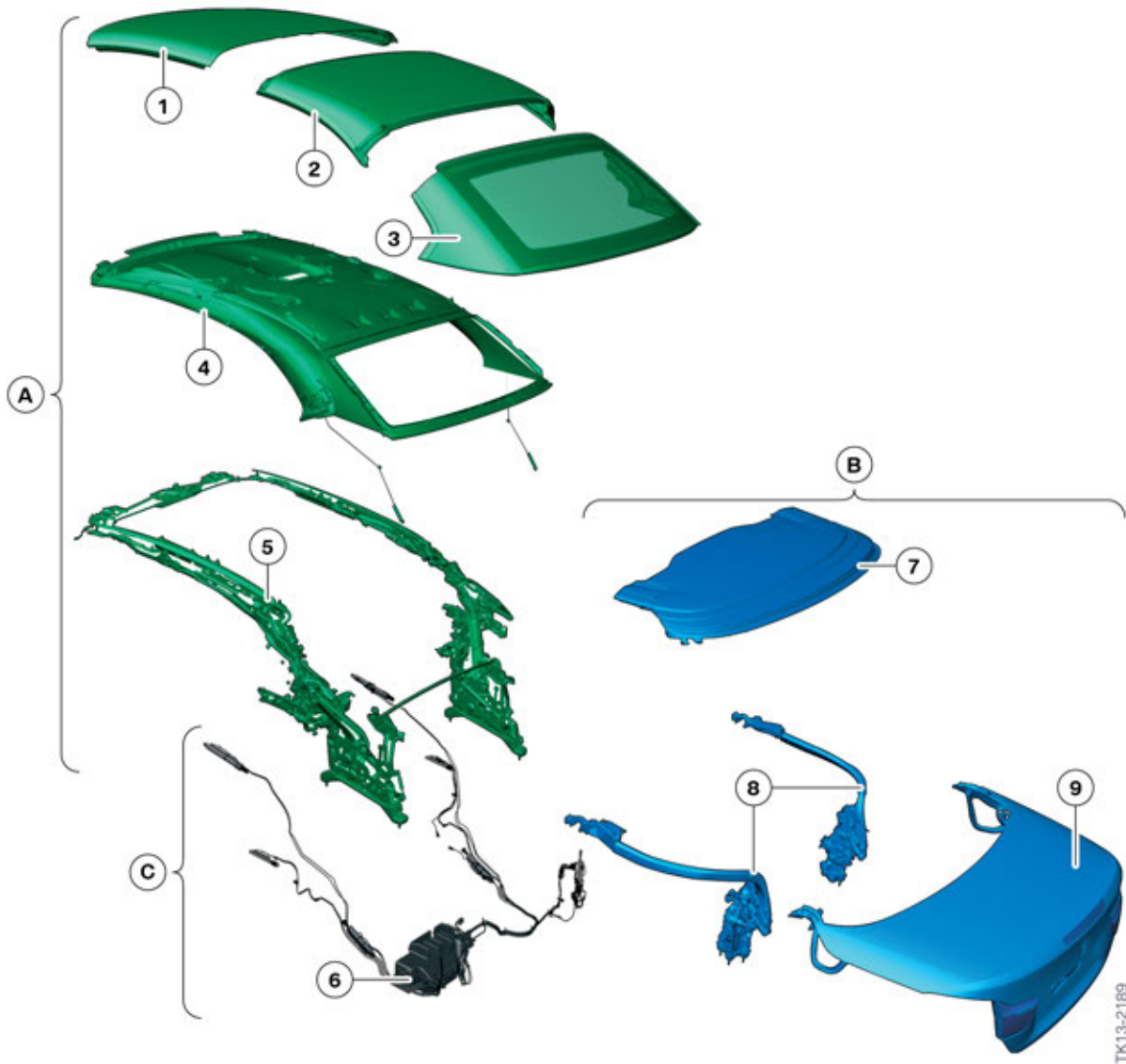
The retractable hardtop weighs about 153 kg (337 lbs) in total with the hydraulic and electrical components. About 102 kg (224 lbs) can be attributed to the roof module and about 51 kg (112 lbs) to the rear module.

F33 Complete Vehicle

6. Retractable Hardtop

6.1. System components

6.1.1. Overview



F33 Structure of the retractable hardtop

Index	Explanation
A	Roof module
B	Rear module
C	Hydraulic system
1	Roof shell, front
2	Roof shell, center

F33 Complete Vehicle

6. Retractable Hardtop

Index	Explanation
3	Roof shell, rear
4	Headliner
5	Roof module linkage (including locking of roof shells)
6	Hydraulic system
7	Hardtop cover
8	Rear module linkage
9	Trunk

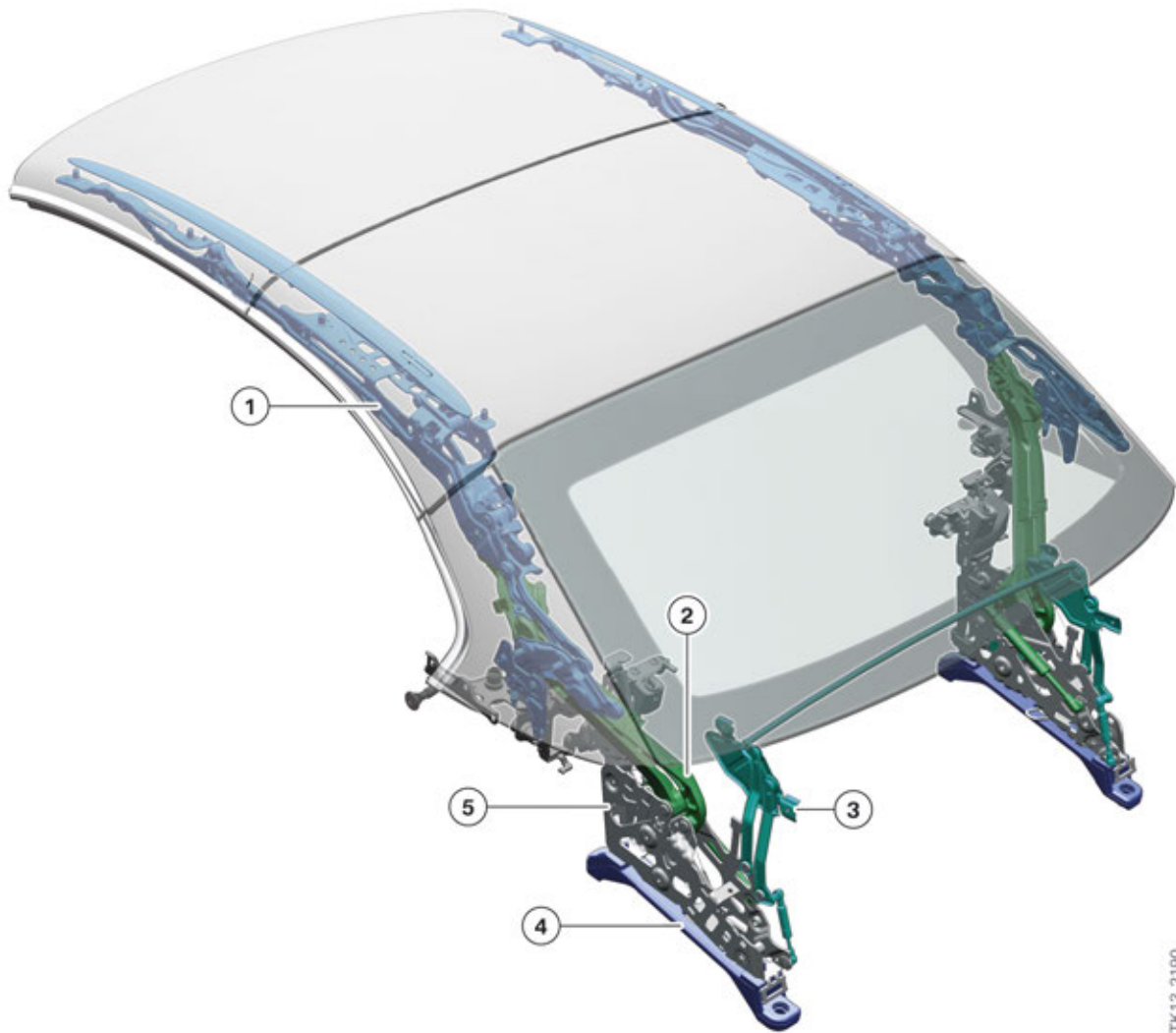
The main retractable hardtop components, divided into mechanical, hydraulic and electrical components, are described in the following text.

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6. Retractable Hardtop

6.1.2. Mechanical components

Kinematics of roof module



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F33 Kinematics of roof module

Index	Explanation
1	Roof module linkage
2	Main pillar
3	Linkage, luggage compartment partition
4	Mounting plates
5	Main bearing

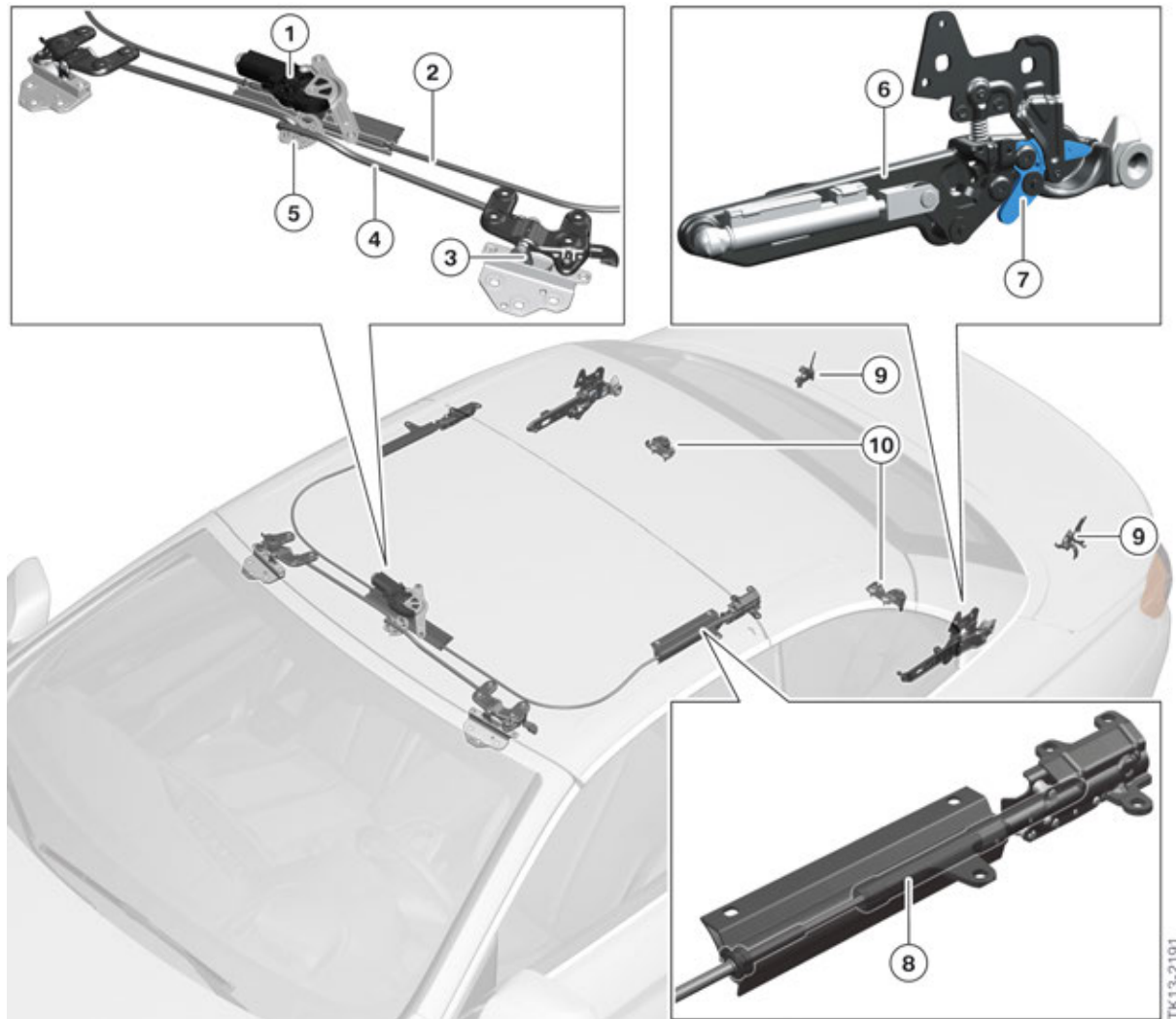
The roof module consists of roof shells, the main bearings, as well as the linkages for the roof module and the luggage compartment partition. The roof module is secured via two mounting plates at the body.

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6. Retractable Hardtop

Compared to the E93, the F33 has a larger crank for the rear roof shell control levers. The roof strip hinges have also been optimized for the F33.

Locking of roof shells



F33 Locking of roof shells

Index	Explanation
1	Electric motor for locking the front roof shell
2	Transmission cable
3	Retaining hook
4	Pushrod
5	Control disc
6	Coupling lock

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6. Retractable Hardtop

Index	Explanation
7	Locking hook
8	Retaining bolt
9	Lock for the main bearing of rear module
10	Lock for the main bearing of roof module

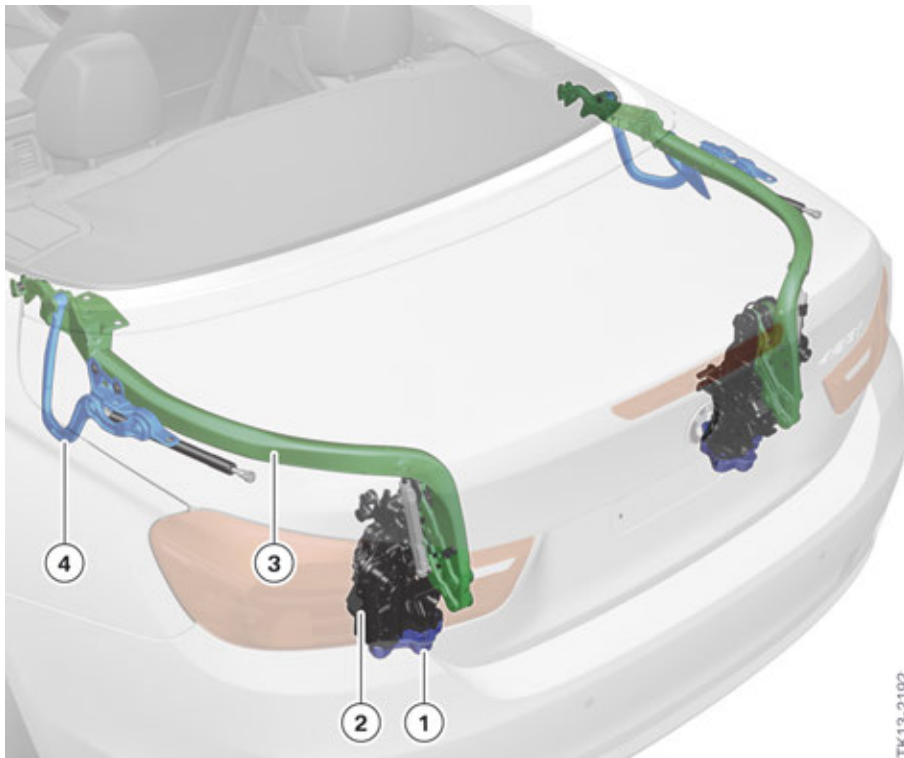
The front roof shell is locked at the front and rear when the hardtop is open and closed. When the hardtop is closed it is locked with the cowl panel and the center roof panel, when the hardtop is open it is locked with the main bearings of the rear and roof modules.

The front roof shell is locked at the front using two retaining hooks and at the rear side using two retaining bolts. An electric motor powers the retaining hooks via a control disk and two pushrods. At the same time, the same electric motor powers the retaining bolts via two transmission cables.

The rear roof shell is locked when the hardtop is closed with the coupling lock. The two locking hooks are powered by a hydraulic cylinder via a relay lever.

The central locking of the front roof shell used in the E93, as well as the locks of the center roof shell and the rear roof shell, are deleted in the F33.

Kinematics of rear module



F33 Kinematics of rear module

F33 Complete Vehicle

6. Retractable Hardtop

Index	Explanation
1	Console
2	Main bearing
3	Linkage (rear module)
4	Linkage (trunk)

The rear module consists of the trunk, hardtop flap, main bearings and linkage. It is secured via two consoles at the body.

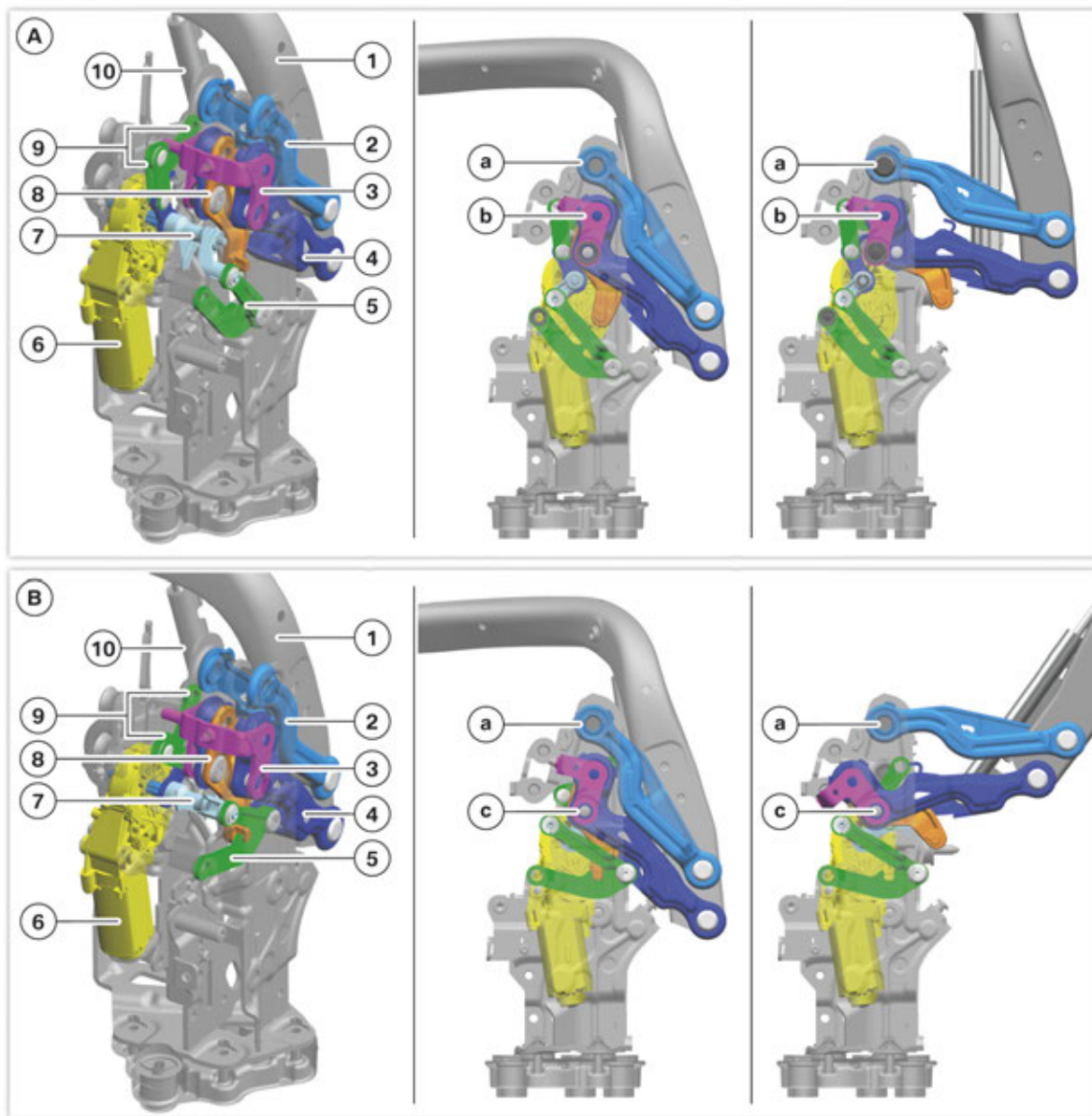
Using electrical kinematics in the two main bearings, two different movements can be performed, powered by the same hydraulic cylinder:

- Move to the rear to move the roof package in or out of the luggage compartment.
- Move to the top for convenient access to the luggage compartment (loading aid).

The kinematics in the two main bearings are powered by the servomotors. The left servomotor is always activated first, and only when the changeover is completed, is the right servomotor activated. After the loading aid is lowered the kinematics switches back to the basic setting "Open/Close hardtop".

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6. Retractable Hardtop



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F33 Main bearing, rear module

Index	Explanation
A	Main bearing, rear module in "Loading aid" position
B	Main bearing, rear module in "Open/Close hardtop" position
a	Center of rotation (control lever up, cannot be adjusted)
b	Center of rotation (variable lever, loading aid position)
c	Center of rotation (variable lever, open/close hardtop position)
1	Support tube
2	Control lever, up

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6. Retractable Hardtop

Index	Explanation
3	Locking lever (loading aid center of rotation)
4	Variable lever
5	Relay lever (hardtop center of rotation)
6	Electric motor
7	Drive arm
8	Locking lever (hardtop center of rotation)
9	Relay lever (loading aid center of rotation)
10	Hydraulic cylinder, rear module

Changeover from loading aid position to open/close hardtop position: The electric drive (6) rotates the two drive arms (7) via a shaft. The inner drive arm (7) retains the locking lever (3) via the relay lever (5). At the same time, the outer drive arm (7) releases the locking lever (9) via the relay lever (8).

The lower center of rotation (c) is now activated. If the hydraulic cylinder extends the rear module, the rear module moves back and the roof package can move in or out of the luggage compartment.

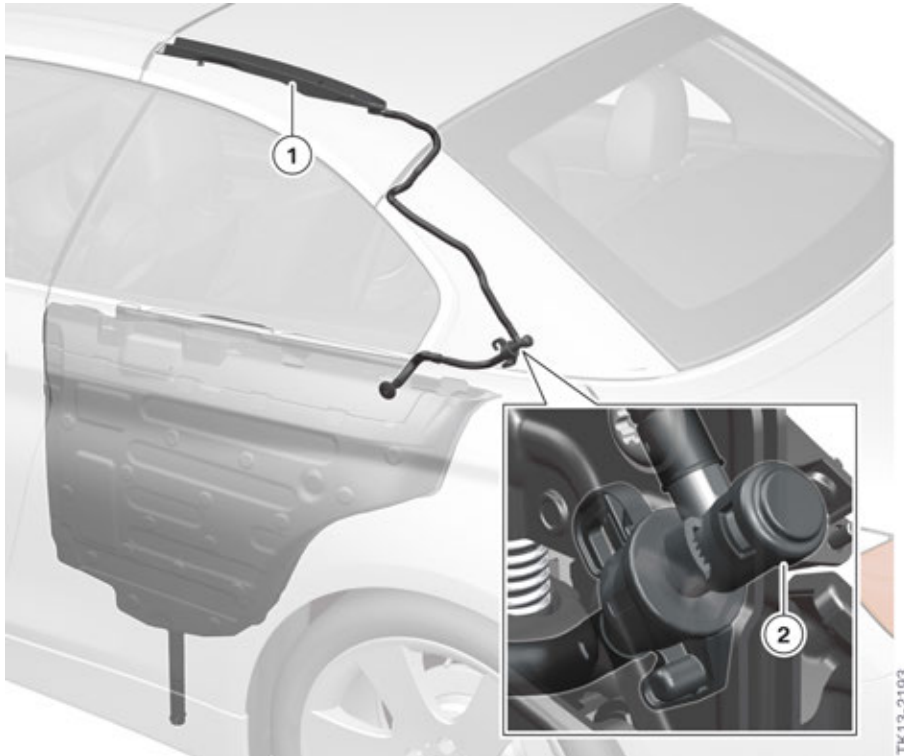
Changeover from open/close hardtop position to loading aid position: The electric drive (6) rotates the two drive arms (7) via a shaft. The inner drive arm (7) releases the locking lever (3) via the relay lever (5). At the same time, the outer drive arm (7) retains the locking lever (9) via the relay lever (8).

The upper center of rotation (b) is now activated. If the hydraulic cylinder extends the rear module, the rear module moves up for convenient access to the luggage compartment.

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6. Retractable Hardtop

Water management



F33 Water management

Index	Explanation
1	Roof duct
2	Valve

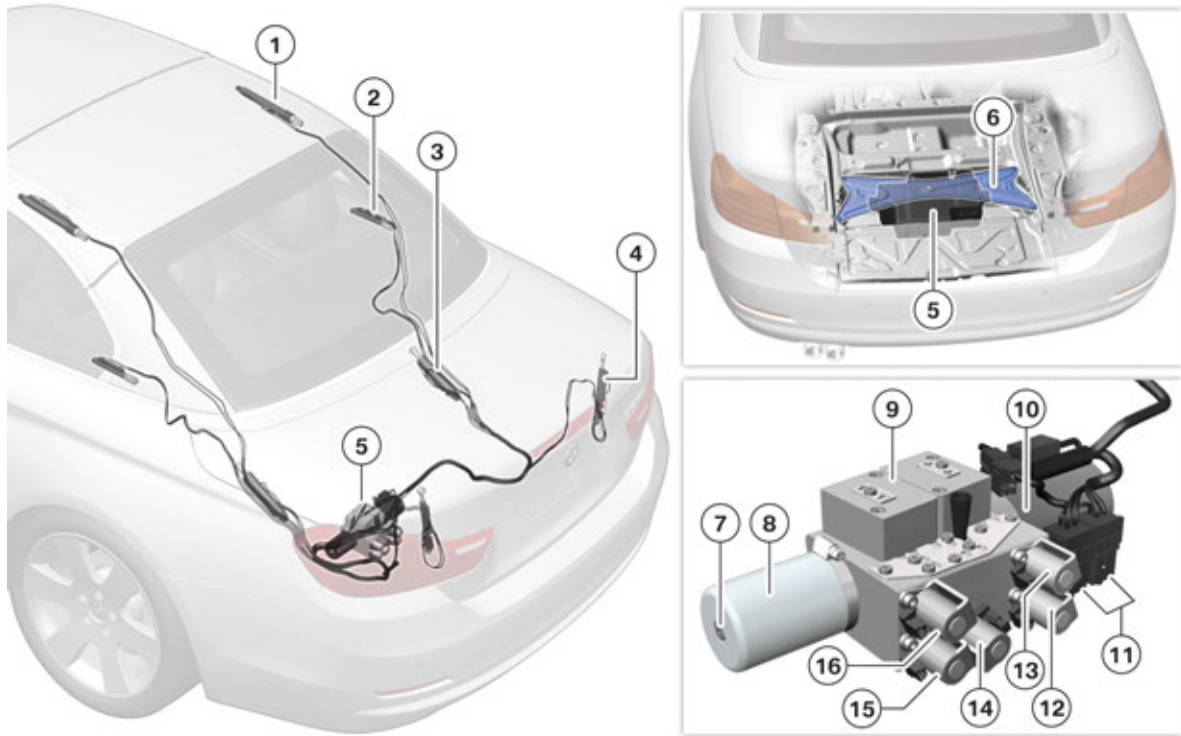
When the hardtop is closed the water from the roof duct runs via a pipe system with valve into a water channel. When the hardtop is open the valve is closed so that any remaining water cannot flow into the luggage compartment.

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6. Retractable Hardtop

6.1.3. Hydraulic components

Layout and function



F33 Overview of the hydraulic components

Index	Explanation
1	Hydraulic cylinder, roof shell
2	Hydraulic cylinder, coupling lock
3	Hydraulic cylinder, main pillar
4	Hydraulic cylinder, rear module
5	Hydraulic unit
6	Reinforcement plate
7	Oil filler plug
8	Oil reservoir for hydraulic fluid
9	Switch block
10	Drive unit (pump motor with hydraulic pump)
11	Relay for activation of the pump motor (2x)
12	Solenoid valve for activating changeover valve 5 (open rear module)

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6. Retractable Hardtop

Index	Explanation
13	Solenoid valve for activating changeover valve 4 (close rear module; lock coupling lock)
14	Solenoid valve for activating changeover valve 3 (unlock coupling lock)
15	Solenoid valve for activating changeover valve 2 (open rear module; unlock coupling lock)
16	Solenoid valve for activating changeover valve 1 (move roof shells apart)

The roof and rear module are operated hydraulically. The hydraulic system consists of eight hydraulic cylinders which are activated via hydraulic lines in pairs by the hydraulic unit. When the hardtop is closed all hydraulic cylinders are retracted. The hydraulic cylinders can be pressurized at the piston and rod ends.

The operating directions of the hydraulic cylinders for the roof shell and hydraulic cylinder for main pillar are dependent on the direction of rotation of the hydraulic pump. The operating directions of the hydraulic cylinder for the coupling lock and the hydraulic cylinder for the rear module are dependent on the position of the corresponding changeover valves.

The hydraulic unit is located in a trough in the luggage compartment floor. In contrast to the E93, in the F33 for the removal of the hydraulic unit an additional reinforcement plate has to be removed. The hydraulic fluid does not have to be changed (long-term oil filling). When topping up the hydraulic fluid, e.g. due to a leak, the hydraulic unit must be removed from the foam jacket. The hydraulic system is automatically bled in the oil reservoir.

The temperature of the drive unit is monitored using a NTC sensor at the electric motor and transmitted to the convertible-top module (CTM). From a temperature of about 95 °C (203 °F) it is no longer possible to open the retractable hardtop. From a temperature of about 105 °C (221 °F) it is also no longer possible to close the retractable hardtop.

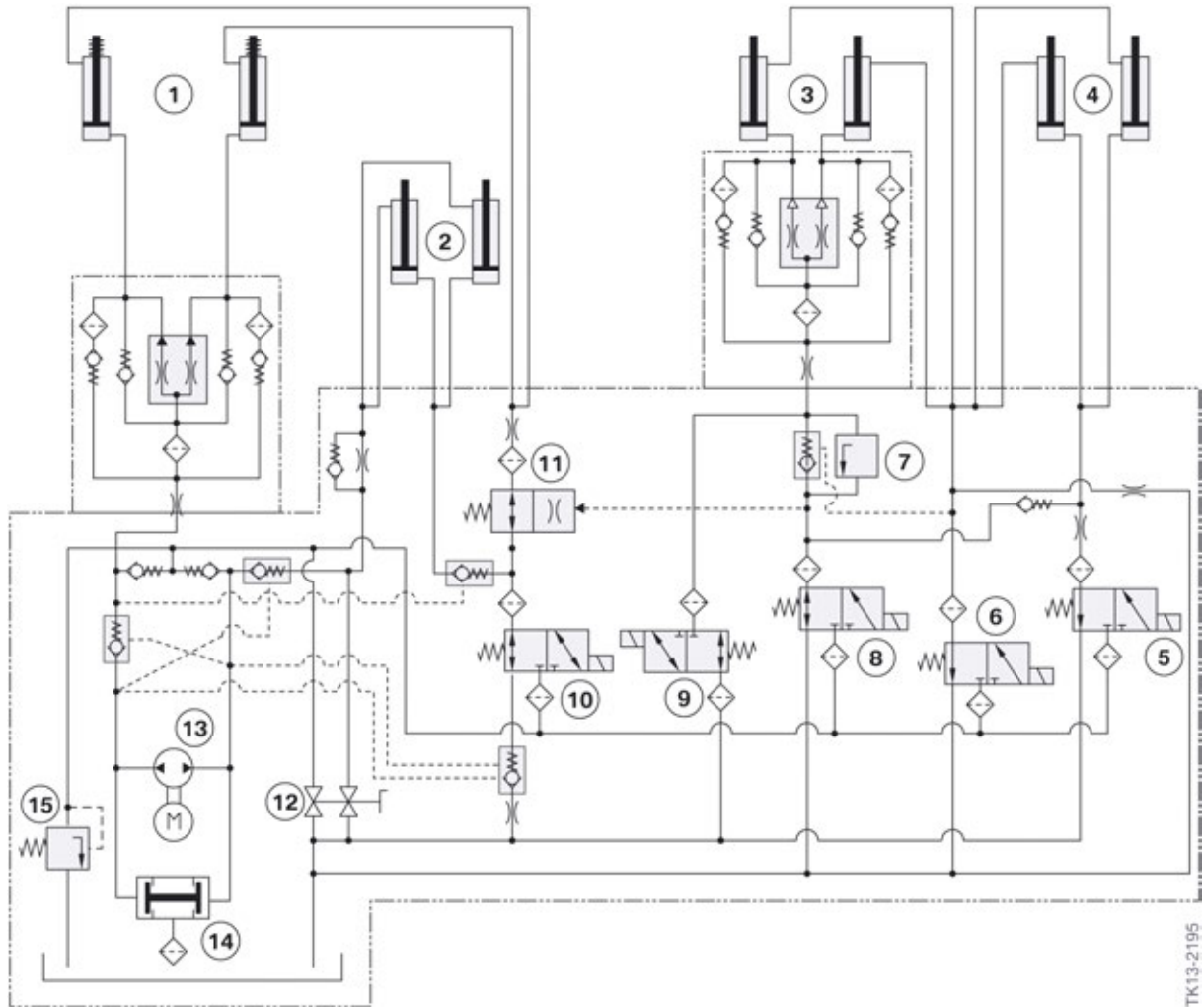
The system pressure is restricted by a pressure limiting valve to 200 bar (+ 10 bar). The operating pressure is maximum 180 bar at room temperature.

A bypass in the hydraulic unit ensures that with the loading aid function the hydraulic cylinder of the rear module is not activated at the full operating pressure. The loading aid can thus be raised or lowered more smoothly. At low ambient temperatures the viscosity of the hydraulic fluid increases and the movement processes would slow down even further. This is why the bypass is closed at ambient temperatures below 10 °C (50 °F) via a bypass valve.

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6. Retractable Hardtop

System wiring diagram for hydraulics of retractable hardtop



F33 System wiring diagram for hydraulics of retractable hardtop

Index	Explanation
1	Hydraulic cylinder, roof shell
2	Hydraulic cylinder, main pillar
3	Hydraulic cylinder, rear module
4	Hydraulic cylinder, coupling lock
5	Changeover valve 3 (unlock coupling lock)
6	Changeover valve 4 (close rear module; lock coupling lock)
7	Pressure-limiting valve
8	Changeover valve 2 (open rear module; unlock coupling lock)
9	Changeover valve 5 (open rear module)
10	Changeover valve 1 (move roof shells apart)

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6. Retractable Hardtop

Index	Explanation
11	Changeover throttle valve
12	Shutoff valve
13	Hydraulic pump
14	Twin pressure valve
15	Pressure-limiting valve

6.1.4. Electrical components

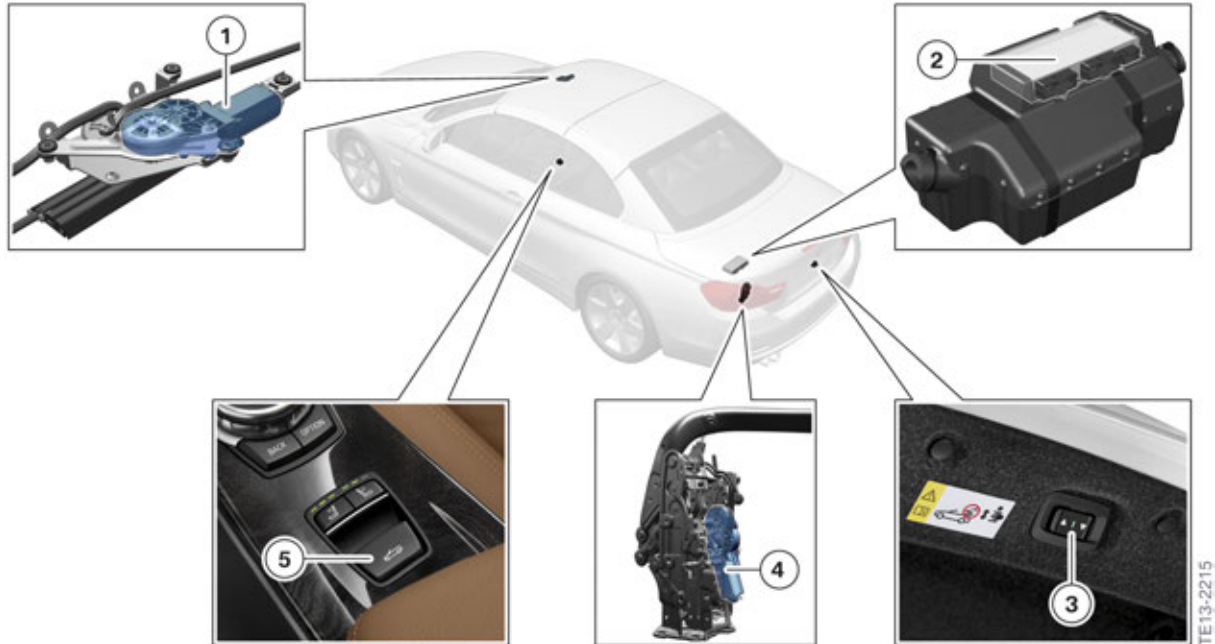
The following electrical components are involved in the function of the retractable hardtop:

- Button for retractable hardtop
- Rocker switch for loading aid
- Interior light of the center roof shell
- Convertible-top module (CTM)
- Pump motor with temperature sensor
- 2 relays for activation of the pump motor
- 5 solenoid valves for activating the changeover valves
- Electric motor for locking the front roof shell
- 2 servomotors for the changeover of loading aid
- 10 microswitches and 10 hall effect sensors for detecting the position of the retractable hardtop

F33 Complete Vehicle

6. Retractable Hardtop

Buttons, control unit and drives



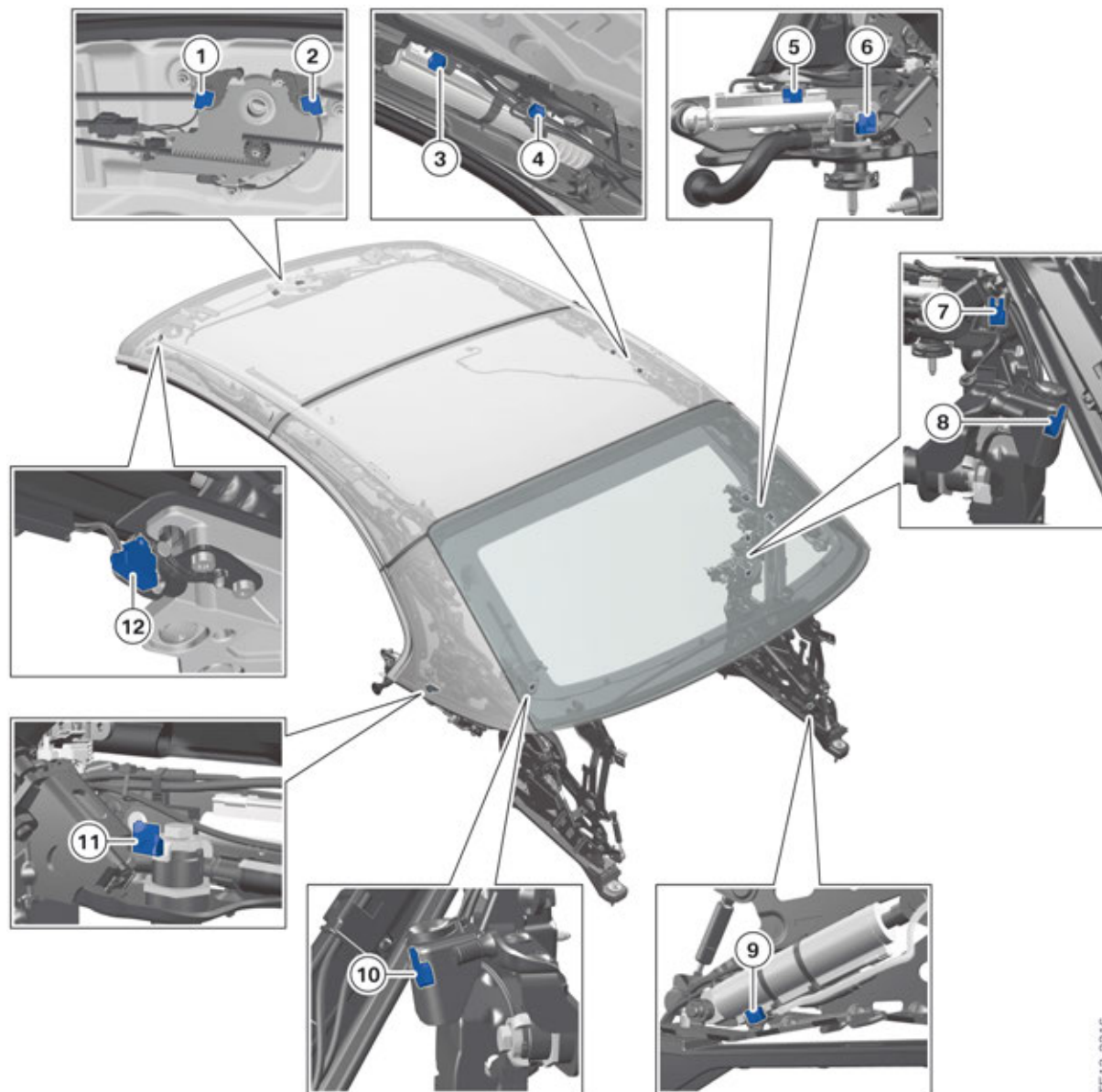
F33 Buttons, control unit and drives of the retractable hardtop

Index	Explanation
1	Electric motor for locking the front roof shell
2	Convertible-top module (CTM)
3	Rocker switch for loading aid
4	Servomotor for changeover of loading aid
5	Button for retractable hardtop

F33 Complete Vehicle

6. Retractable Hardtop

Sensors



TE13-2216

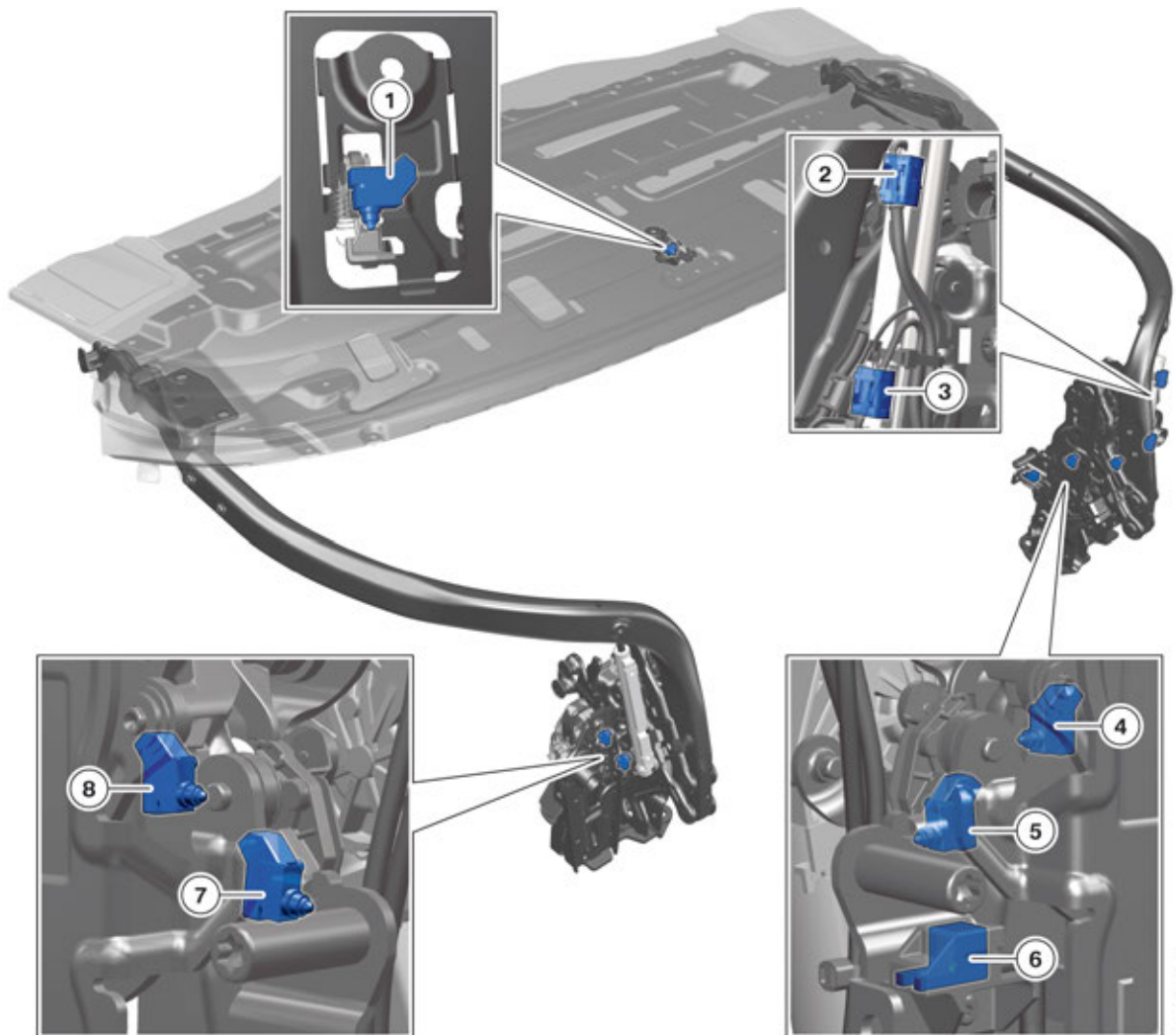
F33 Sensors of roof module

Index	Explanation
1	Microswitch cowl panel locked
2	Microswitch cowl panel unlocked
3	Hall effect sensors of roof shells partially closed
4	Hall effect sensors of roof shells partially open
5	Hall effect sensor for right coupling lock open
6	Microswitch for right coupling lock closed

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6. Retractable Hardtop

Index	Explanation
7	Hall effect sensor for rear module closed
8	Hall effect sensor for roof package stowed positioned, right
9	Hall effect sensor for roof package extended positioned
10	Hall effect sensor for roof package stowed positioned, left
11	Microswitch for left coupling lock closed
12	Microswitch at cowl panel reached



F33 Sensors of rear module

TE13-2217

F33 Complete Vehicle

6. Retractable Hardtop

Index	Explanation
1	Microswitch for retractable hardtop in loading position
2	Hall effect sensor for rear module open
3	Hall effect sensor for damping action of rear module when closing
4	Microswitch for changeover of loading aid on right closed
5	Microswitch for changeover of loading aid on right open
6	Hall effect sensor for luggage compartment partition, bottom
7	Microswitch for changeover of loading aid on left open
8	Microswitch for changeover of loading aid on left closed

All positions of the retractable hardtop which are relevant for the movement process are detected by 10 hall effect sensors and 10 microswitches and sent to the convertible-top module (CTM). All sensors of the retractable hardtop are supplied with voltage by the convertible-top module and have diagnostic capability.

- **Microswitch at cowl panel reached:** The microswitch is located in the cowl panel to the left. If the front roof shell has reached the cowl panel, the microswitch is operated via a relay lever by the centring pin at the front roof shell.
- **Microswitch at front roof shell locked:** The microswitch is located on the left at the carrier plate of the drive for locking the front roof shell. If the retaining hooks and retaining bolts are fully locked, the microswitch is operated by the control disc.
- **Microswitch at front roof shell unlocked:** The microswitch is located on the right at the carrier plate of the drive for locking the front roof shell. If the retaining hooks and retaining bolts are fully unlocked, the microswitch is operated by the control disc.
- **Hall effect sensor of roof shells stowed:** The hall effect sensor is located in the front area of the right hydraulic cylinder of the roof shell. If the roof package is completed stowed, the hall effect sensor is operated.
- **Hall effect sensor for damping action of roof shells when closing:** The hall effect sensor is located in the rear area of the right hydraulic cylinder of the roof shell. It is operated during the closing action shortly before the roof package is completely unpacked. After the hall effect sensor was reached, the front roof shell is moved to the cowl panel with a damping action.
- **Hall effect sensor for coupling lock on right open:** The hall effect sensor is located at the right hydraulic cylinder coupling lock. It is operated when the right coupling lock is completely open. In this position the lock of the rear module and the rear roof shell with coupling lock is removed.
- **Microswitch for right coupling lock closed:** The microswitch is located at the carrier plate of the right coupling lock. If the coupling lock is fully closed, the microswitch is operated via a relay lever.
- **Microswitch for left coupling lock closed:** The microswitch is located at the carrier plate of the left coupling lock. If the coupling lock is fully closed, the microswitch is operated via a relay lever.
- **Hall effect sensor for roof package stowed on right:** The hall effect sensor is located at the right main bearing of the roof module. If the roof package is set down in the luggage compartment, the hall effect sensor is operated by the right centring pin at the front roof shell.

F33 Complete Vehicle

6. Retractable Hardtop

- **Hall effect sensor for roof package stowed on left:** The hall effect sensor is located at the left main bearing of the roof module. If the roof package is set down in the luggage compartment, the hall effect sensor is operated by the left centring pin at the front roof shell. This hall effect sensor is new and not used in the E93.
- **Hall effect sensor for roof package extended positioned:** The hall effect sensor is located in the rear area of the right hydraulic cylinder of the main pillar. If the roof package is completely moved to the front, the hall effect sensor is operated.
- **Hall effect sensor for rear module closed:** The hall effect sensor is located at the right main bearing of the roof module. If the rear module is fully closed, the hall effect sensor is operated by a switch plate at the hardtop flap.
- **Hall effect sensor for rear module open:** The hall effect sensor is located in the upper area of the right hydraulic cylinder of the rear module. The hall effect sensor is operated if the rear module is moved back fully or moved up fully (loading aid).
- **Hall effect sensor for damping action of rear module when closing:** The hall effect sensor is located in the lower area of the right hydraulic cylinder of the rear module. It is operated during the closing action shortly before the rear module is fully closed. After the hall effect sensor is reached, the rear module is closed at the end with a damping action.
- **Hall effect sensor for luggage compartment partition, lowered:** The hall effect sensor is located at the front at the right main bearing of the rear module. It is operated if the luggage compartment partition is folded down fully.

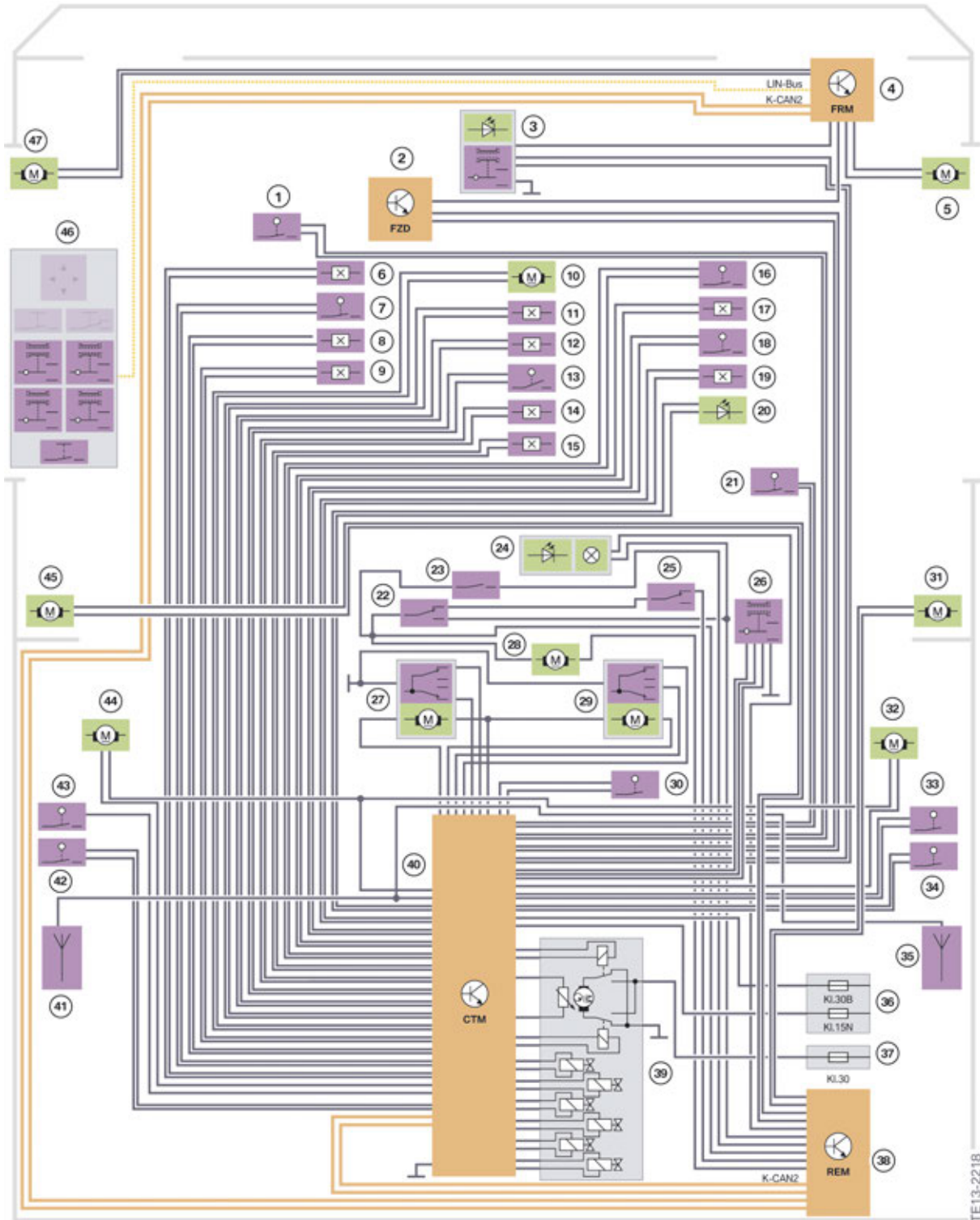
The following sensors are also used for controlling the new loading aid function:

- **Microswitch for retractable hardtop in loading position:** The microswitch is located in the center at the bottom of the hardtop flap. If the roof package is raised fully to the loading position, the microswitch is operated by the rear roof module via a relay lever.
- **Microswitch for changeover of loading aid on right closed:** The microswitch is located in the rear area of the right main bearing of the rear module. If the kinematics is in the opening or closing position of the hardtop, the microswitch is operated.
- **Microswitch for changeover of loading aid on left closed:** The microswitch is located in the rear area of the left main bearing of the rear module. If the kinematics is in the opening or closing position of the hardtop, the microswitch is operated.
- **Microswitch for changeover of loading aid on right open:** The microswitch is located in the front area of the right main bearing of the rear module. If the kinematics is in the position for the loading aid, the microswitch is operated.
- **Microswitch for changeover of loading aid on left open:** The microswitch is located in the front area of the left main bearing of the rear module. If the kinematics is in the position for the loading aid, the microswitch is operated.

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6. Retractable Hardtop

System wiring diagram for convertible-top module (CTM)



F33 System wiring diagram for convertible-top module (CTM)

F33 Complete Vehicle

6. Retractable Hardtop

Index	Explanation
1	Microswitch at cowl panel reached
2	Roof function center (FZD)
3	Button for retractable hardtop
4	Front Electronic Module (FEM)
5	Power window motor, front passenger's side
6	Hall effect sensor for roof package extended positioned
7	Microswitch for left coupling lock closed
8	Hall effect sensor for damping action of rear module when closing
9	Hall effect sensor for rear module open
10	Electric motor for lock at roof shell, front
11	Hall effect sensor for roof package stowed, right
12	Hall effect sensor for roof package stowed, left
13	Microswitch at front roof shell locked
14	Hall effect sensors of roof shells packed
15	Hall effect sensor for damping action of roof shells when closing
16	Microswitch for right coupling lock closed
17	Hall effect sensor for rear module closed
18	Microswitch at front roof shell unlocked
19	Hall effect sensor for coupling lock on right open
20	Interior light of roof shell, center
21	Hall effect sensor for luggage compartment partition, bottom
22	Trunk contact switch, left
23	Trunk push-button
24	Trunk light
25	Trunk contact switch, right
26	Rocker switch for loading aid
27	Soft-close automatic trunk, left
28	Servomotor, trunk lock
29	Soft-close automatic trunk, right
30	Microswitch for retractable hardtop in loading position
31	Power window motor, rear passenger's side
32	Servomotor for changeover of loading aid, right
33	Microswitch for changeover of loading aid on right open
34	Microswitch for changeover of loading aid on right closed
35	Side panel antenna (FM2)

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6. Retractable Hardtop

Index	Explanation
36	Power distribution box, luggage compartment
37	Battery power distribution box
38	Rear Electronic Module (REM)
39	Hydraulic unit
40	Convertible-top module (CTM)
41	Side panel antenna (FM1 and AM)
42	Microswitch for changeover of loading aid on left closed
43	Microswitch for changeover of loading aid on left open
44	Servomotor for changeover of loading aid, left
45	Power window motor, rear driver's side
46	Switch block, driver's door
47	Power window motor, front driver's side

6.2. Operation and function

6.2.1. Retractable hardtop – Standard operation



The hardtop moves back and up during opening and closing. For the movement of the hardtop, for example in garages, ensure that there is at least 2 meters clearance above and at least 50 centimeters at the rear, otherwise it may incur damage.

When the rollover protection is activated do not move the hardtop, otherwise it may cause damage to the hardtop.

Do not open or close the hardtop on uneven terrain, otherwise the vibrations may cause damage to the hardtop.



Danger of injury!

Do not reach into the hardtop mechanism during the opening and closing action. Always keep a safe distance from the movement range of the hardtop.

Standard operation of the retractable hardtop is performed using the retractable hardtop button in the center console.

F33 Complete Vehicle

6. Retractable Hardtop



F33 Button for retractable hardtop

Movement of the retractable hardtop is performed as long as the button is operated.

- Press button for the retractable hardtop: Hardtop is open
- Pull button for the retractable hardtop: Hardtop is closed

If the button for the retractable hardtop is released during opening or closing of the hardtop, the movements of the hardtop or the side window are interrupted immediately and a Check Control message is displayed in the instrument cluster. The movements are resumed when the retractable hardtop button is operated again.

Prerequisites

To open or close the retractable hardtop using the button in the center console, the following preconditions must be satisfied:

Open retractable hardtop	Close retractable hardtop
Terminal R or terminal 15N activated	Terminal R or terminal 15N activated
Ambient temperature > -10 °C (14 °F)	
Trunk closed	Trunk closed
Luggage compartment partition engaged in lowermost position	
Loading aid lowered fully	Loading aid lowered fully
Power window regulators are initialized	Power window regulators are initialized
Vehicle voltage > 10.5 V	Vehicle voltage > 10.5 V
Pump motor temperature < 95 °C (203 °F)	Pump motor temperature < 105 °C (221 °F)
Max. driving speed 13 km/h* (8 mph*)	Max. driving speed 13 km/h* (8 mph*)
Roll tendency/lateral acceleration of the vehicle not too high	Roll tendency/lateral acceleration of the vehicle not too high

* A movement which has already started is continued up to a driving speed of 18 km/h (11 mph).

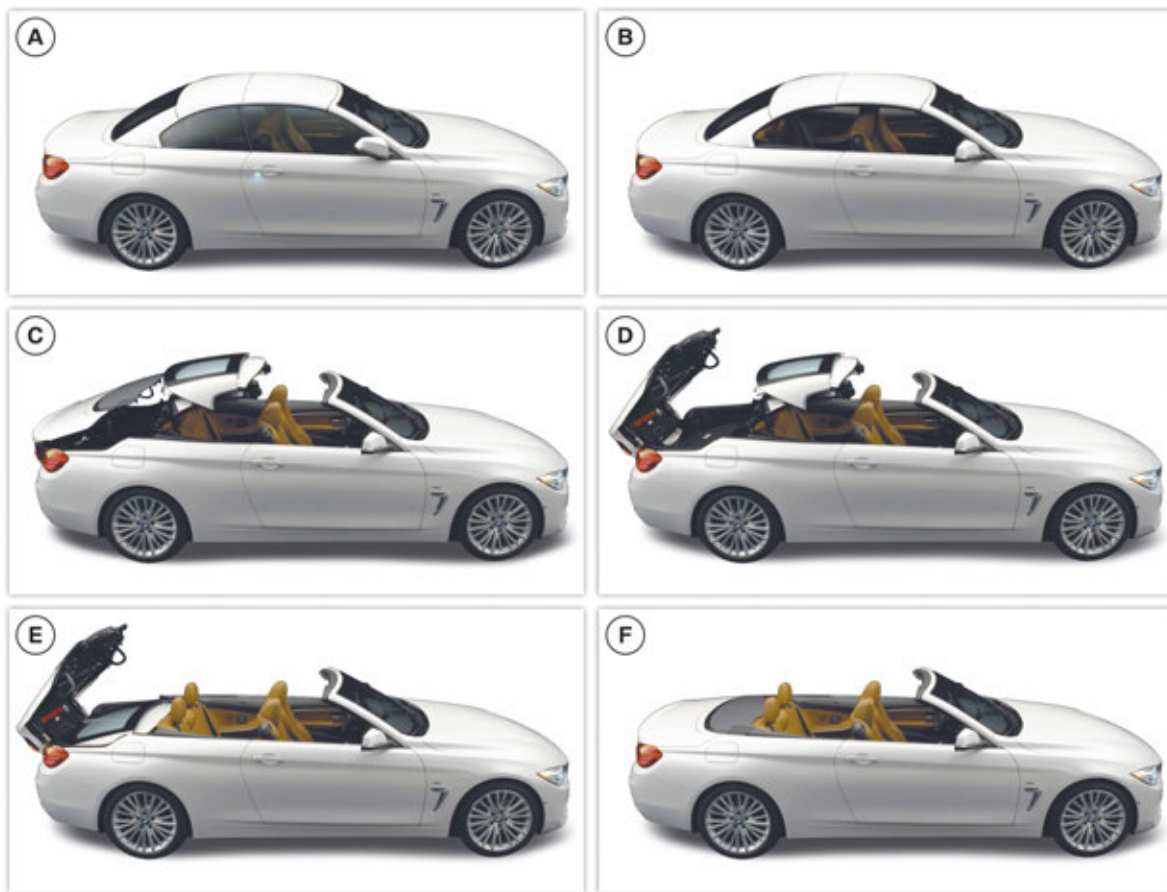
F33 Complete Vehicle

6. Retractable Hardtop

If the retractable hardtop button is operated and one of the preconditions is not satisfied, the opening or closing action is not started. In addition, a Check Control message is displayed in the instrument panel.

Process for opening the retractable hardtop

By pulling the retractable hardtop button the preconditions for opening the retractable hardtop are checked by the convertible-top module (CTM). If all the preconditions are satisfied, the retractable hardtop is opened.



TK13-2196

F33 Opening the retractable hardtop

F33 Complete Vehicle

6. Retractable Hardtop

Index	Explanation
A	The retractable hardtop and the side windows are closed.
B	The side windows are lowered completely. The front roof shell is unlocked at the cowl panel and at the center roof shell. The rear roof shell and the rear module are unlocked at the coupling locks.
C	The roof shells slide over each other. The rear module starts to open.
D	The rear module is opened fully.
E	The roof package is set down. The front roof shell is locked at the main bearings of the rear and roof modules.
F	The rear module is closed and locked at the coupling locks (shortly before the rear module is closed fully, the movement is slowed down slightly).

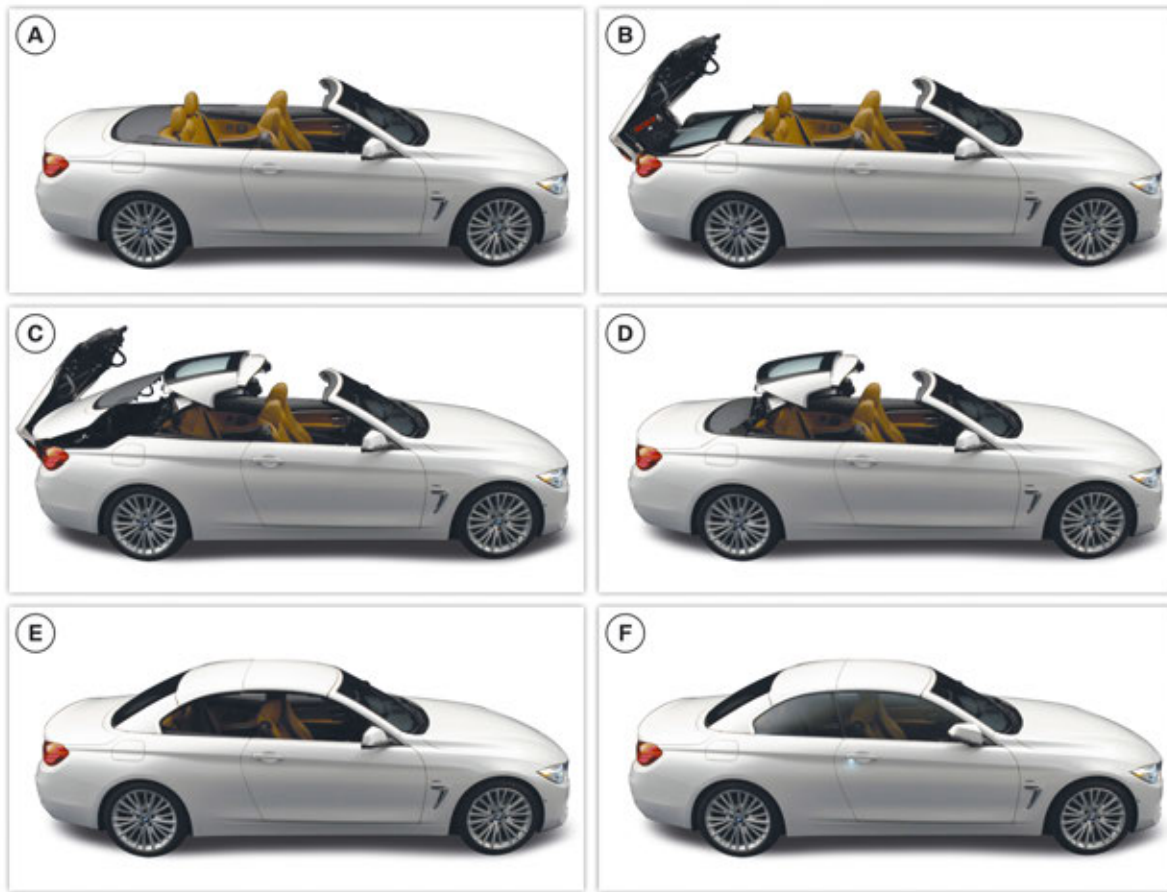
After the hardtop is opened fully, the retractable hardtop button can be held or pulled again for a brief period to raise the side windows up again.

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6. Retractable Hardtop

Process for closing the retractable hardtop

By pressing the retractable hardtop button the preconditions for closing the retractable hardtop are checked by the convertible-top module (CTM). If all the preconditions are satisfied, the retractable hardtop is closed.



TK13-2197

F33 Closing the retractable hardtop

Index	Explanation
A	The side windows are lowered completely.
B	The rear module is unlocked at the coupling locks and opened fully. The front roof shell is unlocked at the main bearings of the rear and roof modules.
C	The roof package is positioned.
D	The rear module is closed and locked at the coupling locks and unlocked again (shortly before the rear module is closed fully, the movement is slowed down slightly).
E	The roof shells are moved apart from each other (shortly before the front roof shell reaches the cowl panel, the movement is slowed down slightly).
F	The front roof shell is locked at the cowl panel and at the center roof shell. The rear roof shell and the rear module are locked at the coupling locks. The side windows are raised completely.

F33 Complete Vehicle

6. Retractable Hardtop

6.2.2. Retractable hardtop – Convenience functions

In vehicles equipped with the optional equipment Comfort Access (SA 322) the retractable hardtop can also be opened and closed with the ID transmitter, and closed via the outside door handle electronics.

For the convenience function of the retractable hardtop terminal R or terminal 15N cannot be activated. The ID transmitter must however be situated in the immediate vicinity of the vehicle. Otherwise the same prerequisites as for standard operation apply.

Convenient opening and closing using the ID transmitter

To open or close the retractable hardtop with the ID transmitter, the Unlocking/Locking button must be pressed and held down. After a short waiting period the retractable hardtop is opened or closed. The opening or closing action is immediately interrupted when the Unlocking/Locking button is released or the button on a different ID transmitter is pressed or the close-range area is left.

Convenience closing via outside door handle electronics

The sensitive surface of the outside door handle must be touched in order to close the retractable hardtop. After a short waiting period the closing action of the retractable hardtop is started. The sensitive surface must be constantly touched, otherwise the closing action will be immediately interrupted.

6.2.3. Loading aid



Before operating the loading aid, ensure that the movement area of the loading aid is free. Otherwise, damage may occur.



Danger of injury!

Do not reach into the mechanism during lowering or raising of the loading aid. Always keep a safe distance from the movement area of the loading aid.

F33 Complete Vehicle

6. Retractable Hardtop



TK13-2198

F33 Loading aid

Index	Explanation
A	Rear module lowered
B	Rear module raised (loading aid)
1	Rocker switch for loading aid

The newly developed, standard equipment loading aid enables convenient loading of the luggage compartment when the hardtop is open. The entire hardtop together with the roof package, trunk and hardtop flap is raised about 40 cm (15.7 in) hydraulically.

For the loading aid function the kinematics in the main bearings of the rear module is switched electrically.

The operation is engaged via a rocker switch at the lower edge of the trunk. The following conditions are shown by a LED in the rocker switch:

- LED illuminates: Loading aid in respective end position
- LED flashes quickly: Loading aid not in end position
- LED flashes slowly: Prerequisites not satisfied

F33 Complete Vehicle

6. Retractable Hardtop

Prerequisites

In order to raise or lower the loading aid, the following preconditions must be satisfied:

- Vehicle is on an even surface
- Retractable hardtop is fully opened
- Trunk fully open
- Luggage compartment partition engaged in lowermost position
- Ambient temperature > -10 °C (14 °F)

After the ignition is switched off the raised loading aid can be lowered within about 10 minutes. The ignition must be switched back on again to lower the loading aid.

6.3. Notes for Service



The repair instructions for repairs to the hardtop must be observed.

When repairing the roof shells always remove and install the roof shells one at a time.

For troubleshooting at the retractable hardtop also observe the non-electrical diagnosis (NED) in the diagnosis system (ISTA).

For repairs to damaged vehicles it is necessary to follow the notes in the diagnosis system (ISTA) for the procedure at the retractable hardtop.

If the open retractable hardtop or the raised loading aid are defective, they can be closed only by BMW Service using special tools. The emergency closing operation was not designed to be performed by the customer.

Due to the additional loading aid function the process for the emergency closing has changed in comparison to the E93, see repair instructions for more information.

After the removal and installation of the servomotors for the changeover of the loading aid these must be initialized using a service function in the diagnosis system (ISTA).

6.3.1. Check Control messages

In certain situations it will not be possible to open or close the retractable hardtop. In this case, one or more Check Control messages are displayed in the instrument cluster and appear as plain text in the central information display.

F33 Complete Vehicle

6. Retractable Hardtop

The following table provides an overview of the possible messages:

Check Control symbol	Check Control message	Text in central information display
	Loading aid, Stop with care	Loading aid Lower loading aid before start of the journey. See Owner's Manual.
	Roof. Incorrect loading of luggage compartment	Roof Close convertible roof fully. Check luggage compartment for incorrect loading, see Owner's Manual.
	Roof, Drive more slowly	Roof Speed for opening/closing the roof exceeded. Reduce speed and continue opening/closing.
	Roof movement complete	
	Luggage compartment partition	Luggage compartment partition Put luggage compartment partition in lower position.
	Roof. Only at standstill	Roof Only possible to open/close roof when vehicle is at a standstill.
	Roof	Roof Opening/closing temporarily not possible. Possible causes, see Owner's Manual.
	Roof not locked!	Roof Roof not locked. Fully open or close roof first, then continue driving. See Owner's Manual.

F33 Complete Vehicle

7. Luggage Compartment



F33 Luggage compartment

The luggage compartment capacity of the BMW 4-Series Convertible is 370 liters (13.1 ft³) when the hardtop is closed and 220 liters (7.8 ft³) when the hardtop is open. It is thus 20 liters (0.7 ft³) greater when the hardtop is closed and 10 liters greater when the hardtop is open than the predecessor.

A through-loading system (SA 465) is standard equipment along with Split fold-down rear seat. The through-loading opening has been enlarged for more ergonomic loading of large items of luggage. After the privacy shield and the rear seat backrest are folded down, a smooth transition arises and large items of luggage can be pushed through.

To protect the luggage compartment from unauthorized access, the privacy shield can only be opened from the luggage compartment.



Luggage can only be stowed under the luggage compartment partition, as well as when the hardtop is closed in the side areas. Observe corresponding labels. Before opening the hardtop remove all items of luggage from the side areas, otherwise, damage may occur to the luggage and hardtop.



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